JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-05 [P1]

KEY SHEET

CHEMISTRY

1)	**A**	2)	**C**	3)	**D**	4)	**A**	5)	**A**	6)	C	7)	A	8)	C
9)	**ACD**	10)	**AB**	11)	**ABCD**	12)	**BCD**	13)	**5**	14)	7	15)	2	16)	4
17)	**7**	18)	**4**	19)	a-prt	b-pst	c-qs	d-qrst		20)	a-pqs	b-qs	c-q	d-rs	

PHYSICS

21)	**B**	22)	**A**	23)	**C**	24)	**A**	25)	**B**	26)	**A**	27)	**D**	28)	**C**
29)	**ACD**	30)	**ABC**	31)	**AD**	32)	**AB**	33)	**9**	34)	**9**	35)	**1**	36)	**6**
37)	**8**	38)	**3**	39)	A-R	B-QT	C-T	D-S		40)	A-P	B-Q	C-R	D-S	

MATHEMATICS

41)	**C**	42)	**D**	43)	**B**	44)	**B**	45)	**D**	46)	**C**	47)	**B**	48)	**B**
49)	**AB**	50)	**BC**	51)	**AB**	52)	**AB**	53)	**0**	54)	**6**	55)	**5**	56)	**3**
57)	**0**	58)	**7**	59)	**A-Q**	**B-S**	**C-S**	**D-T**		60)	**A-Q**	**B-S**	**C-P**	**D-T**	

SOLUTIONS
CHEMISTRY

1. H_2O $\quad AgBv \rightleftharpoons \underset{S}{Ag^+} + \underset{S}{Br^-} \qquad k \,\&\, p = S^2$

$AgNO_3 \qquad\qquad S_1 + 10^{-7} \;\; S_1$

$S_1\left(S_1 + 10^{-7}\right) = 12\times10^{-14}$

$S_1 = 3\times10^{-7}$

$\therefore \left[Ag^+\right] = 4\times10^{-7}M \left[Br^-\right] = 3\times10^{-7}M$

And $\left[NO_3^-\right] = 10^{-7}M$

$K' = \dfrac{\Lambda_m \times M}{1000}$

$K'_{Ag^+} = \dfrac{4\times10^{-3}\times10^4\times4\times10^{-7}}{1000} = 16\times10^{-9}$

$K'_{B_1^-} = \dfrac{6\times10^{-3}\times10^4\times3\times10^{-7}}{1000} = 18\times10^{-9}$

$K'_{NO_3^-} = \dfrac{5\times10^{-3}\times10^4\times10^{-7}}{1000} = 5\times10^{-9}$

$K'_{overall} = 39\times10^{-9}\,scm^{-1} = 39\times10^{-7}\,sm^{-1}$

2. $A \rightleftharpoons 2B + C$

$t=0$	a	0	0
$t=20$	$a-x_1$	$2x_1$	x_1
$t=30$	$a-x_2$	$2x_2$	x_2
$t=50$	$a-x_3$	$2x_3$	x_3
$t=\infty$	0	$2a$	a

$t = 20\left(a - x_1\right) \propto 30^0$

$t = 50\left(a - x_1\right) \propto 15^0$

$\therefore$ half life =30 min

At t=30 min volume of H_2O_2 consumed in 30ml (50% B)

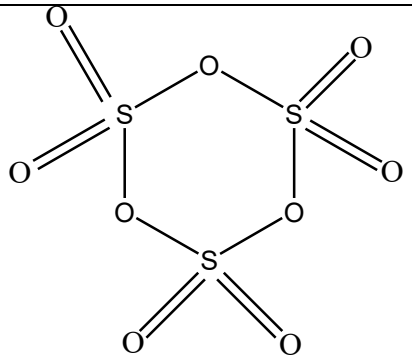
At t=90 min (3half lives) 'B' produced 87.5%

$\therefore V_{H_2O_2} = 60\times\dfrac{87.5}{100} = 52.5$

3. 'U' is visible region then the line belonging to U.V region should have higher energy than U

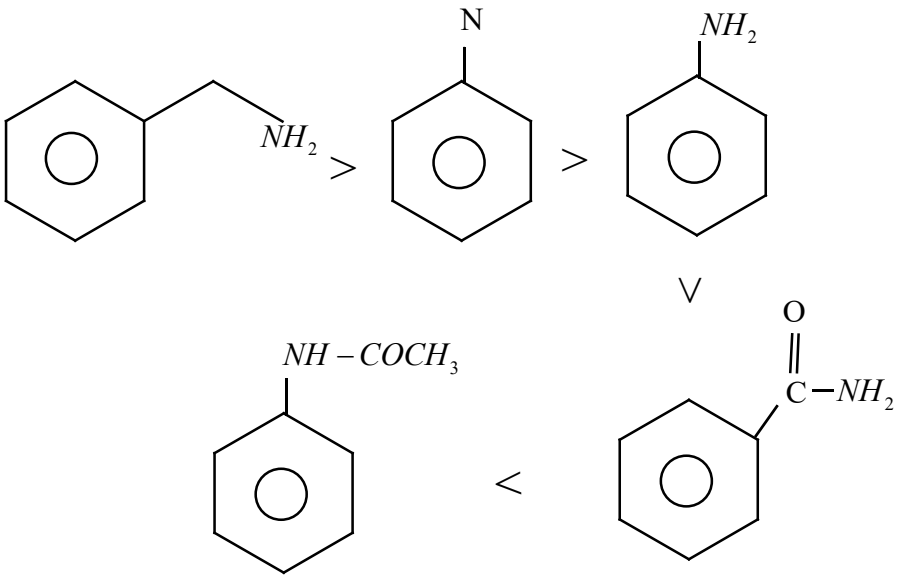
4. Coagulation value α 1/charge on ion

5.



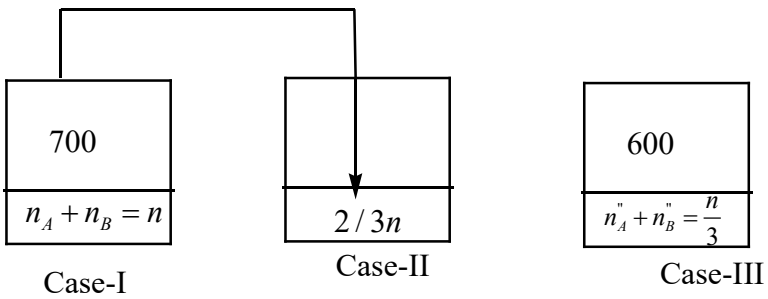
6. Q, & R are same compounds

7.



8. With benzene, decarbonylation of acylium ion occurs and hence alkylation product is formed. Anisole, being more reactive, reacts quickly with acylium ion to give normal acylation product as the major product.

9.



Case-I $p_A^0 \frac{nA}{n} + p_B^0 \frac{nB}{n} = 700$ ….. (1)

$p_A^0\left(\frac{nA}{n}\right) = 700 \times \frac{3}{4}$ $p_B^0\left(\frac{nB}{n}\right) = 700 \times \frac{1}{4}$

Case-II $n''_A = \frac{0.3n}{3}$ $n''_B = \frac{0.7n}{3}$

$P_A^o \frac{n''_A}{n} + P_B^0 \frac{n''_B}{n} = 600$ ….. (2)

Condensate: $\frac{n_A^1}{\frac{2}{3}n} = \frac{3}{4} \Rightarrow n_A^1 = \frac{n}{2}$ $\frac{n_B^1}{\frac{2}{3}n} = \frac{1}{4} \Rightarrow n_B^1 = \frac{n}{6}$

$n_A = n_A^{'} + n_A^{''} = 0.1n + \frac{n}{2} = 0.6n \quad n_B = n - 0.6n = 0.4n$

From 1

$$P_A^o(0.6) + P_B^0(0.4) = 700$$

From 2

$$P_A^o \frac{(0.3)}{3} + P_B^0\left(\frac{0.7}{3}\right) = 600$$

Upon solving $P_A^o = \frac{2500}{3}$

$$P_B^o = 500$$

$X_A = 0.6 \; X_B = 0.4$

10. $NH_4^+ \rightarrow 109.5^o$

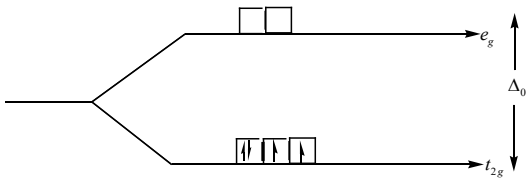
$NH_3 \rightarrow 107^o$

$NH_2^- \rightarrow 104.5^o$

No. of lone pairs in $S_8 = 2 \times 8 = 16$

$BMe_3 \rightarrow$ Triangle planar

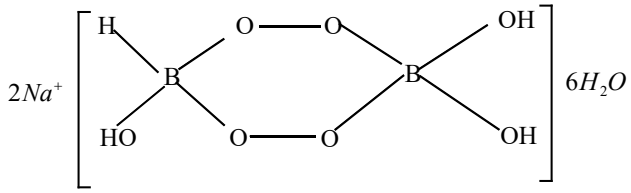
11.



$CFSE = \left[-4(0.4) - 0\right]\Delta_0 + D \qquad = (-1.6 \times 38500) + 28000 = -33600 cm^{-1}$

12. A. $Ca_2B_6O_{11} + Na_2CO_3 \rightarrow Na_2B_4O_7 + CaCO_3 + NaBO_2$

$2NaBO_2 + 2H_2O_2 + 6H_2O$



Sodium perxo borate

$Na_2B_4O_7 + 2HCl \rightarrow 2NaCl + H_2B_4O_7$

HF

HBF_4

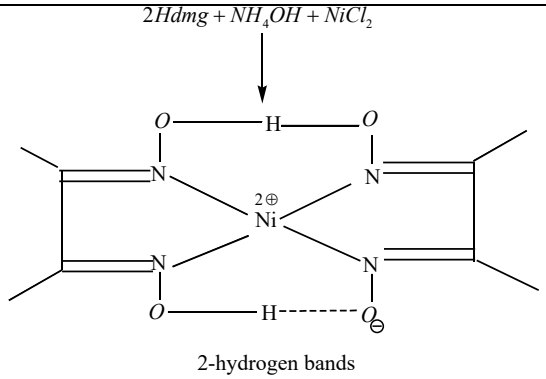
$Cu(BO_2)_2 \rightarrow$ Bheal Bead

13. a,b,c,d,e,f are correct

14. $\underset{(A)}{MnO_2} + KOH + KClO_3 + \underset{(B)}{K_2MnO_4} + KCl$

$K_2MnO_4 + H_2O + CO_2 \rightarrow \underset{(C)}{KMnO_4} + K_2CO_3 + \underset{(A)}{MnO_2}$

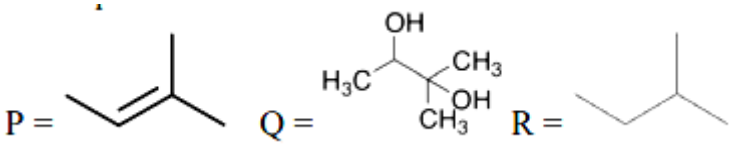
15.



2-hydrogen bands

16. (i) (ii) (iii) (iv) are correct

17.

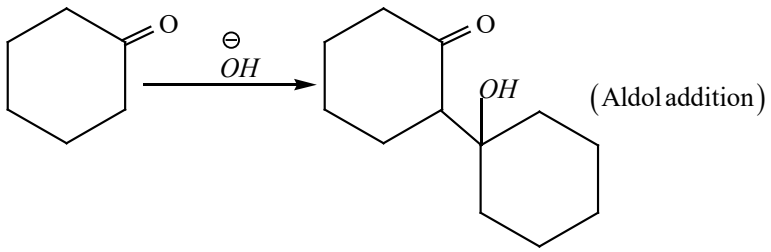


i,ii,iii,iv,vi re correct

18. $x = 10, y = 10, z = 5$

19. a-prt, b-pst, c-qs, d-qrst

20. A) $ph - \underset{\underset{O}{\|}}{C} - H \xrightarrow{\text{alkaline KCN}} ph - \underset{\underset{OH}{|}}{C}H - \underset{\underset{O}{\|}}{C} - Ph$ (Benzoin condensation) (α -hydoroxycarbonyl)



B)

(C) $CH_3-\underset{\underset{O}{\|}}{C}-OEt \xrightarrow[(2)\ H_3O^{\oplus},\ \Delta]{(1)\ C_2H_5ONa} CH_3-\underset{\underset{O}{\|}}{C}-CH_3$ (Claisen ester condensation)

(D) $Ph-CH=O \xrightarrow[(2)\ H^{\oplus}]{(1)\ KOH} Ph-COOH + Ph-CH_2-OH$ (Cannizzaro reaction)

PHYSICS

21. $mg - T = ma$

$T - F = ma \quad F = Bil$

22. $dR = \dfrac{\rho dx}{A}$

23. Use steady state condition

24. $x = \dfrac{f}{\cos\theta} - 1$

25. $\delta_m = 2i - A = A$

$\Rightarrow i_1 = i_2 = A \, and \, r_1 = r_2 = \dfrac{A}{2} \quad \Rightarrow r_1 = \dfrac{i_1}{2}$

Using smell's law $\sin A = \mu \sin\dfrac{A}{2} \qquad \Rightarrow A = 2\cos^{-1}\left(\dfrac{\mu}{2}\right)$

At the second surface $\Rightarrow i_2 = \dfrac{\pi}{2} \Rightarrow r_2 = C$

$\therefore r_1 = A - C \Rightarrow \sin i_1 = \mu \sin(A - C)$

26. $F = \pi R^2 \rho g h - \dfrac{1}{3}\pi R^2 \rho g h = \dfrac{2}{3}\pi R^2 \rho g h$

27. $dI = jdA \qquad \Rightarrow dI = kr^2(2\pi r dr)$

$\Rightarrow I = 2\pi K \int_0^a r^3 dr \Rightarrow I = \dfrac{\pi k a^4}{2}$

Further, field for r>a is $B = \dfrac{\mu_0 i}{2\pi r} = \dfrac{\mu_0 k a^4}{4r}$

And field for r<a is $B = \dfrac{\mu_0\left(\dfrac{\pi k r^4}{2}\right)}{2\pi r} = \dfrac{1}{4}\mu_0 k r^3$

28. In case of pure rolling

$f = \dfrac{mg\sin\theta}{1 + \dfrac{mR^2}{I}}(\text{up ward}) \qquad \therefore f \alpha \sin\theta$

29. $dR = \dfrac{\rho dx}{A}$

30. $L_1 i_1 = L_2 i_2 \, and \, i = i_1 + i_2$

31. $f^{\max} = \mu N = \mu Bil \qquad i = \dfrac{100}{5 + 0.5t}$

32. $|F| = \dfrac{dU}{dr} = \dfrac{ke^2}{r^4}$

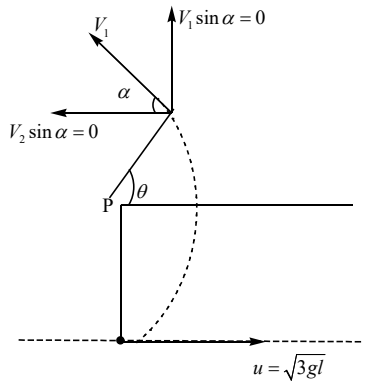
$\dfrac{ke^2}{r^4} = \dfrac{mv2}{r}$ ..... (1)

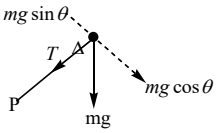
And $mvr = \dfrac{nh}{2\pi}$ (2)

By II and I $\qquad r = \dfrac{ke^2 4\pi m}{n^2 h^2} = k_1 \dfrac{m}{n^2}$ $\qquad$ Total energy $= -\dfrac{1}{2}$ potential energy

$= \dfrac{-ke^2}{6r^3} = \dfrac{-ke^2 n^6}{6k_1^3 m^3} \qquad \therefore$ Total energy $\alpha n^6 \qquad$ Total energy αm^{-3}

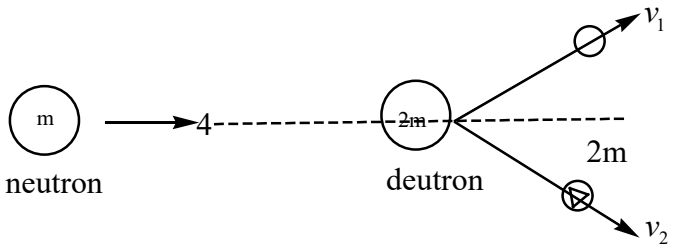
33.





$T + mg\sin\theta = \frac{mv_1^2}{\ell}$ For minimum velocity T=minimum=0

34.



Using linear momentum conservation in Horizontal and vertical condition

$mu = mv_1\cos\theta + 2mu_2\cos\varphi$

$O = mv_1\sin\theta - 2mu_2\sin\varphi$ Also, $\left(\frac{2}{3}\right)\frac{1}{2}mu^2 = \frac{1}{2}\times 2mv_2^2$

$\left(\frac{1}{3}\right)\frac{1}{2}mu^2 = \frac{1}{2}\times 2m\times v_1^2$

35. $A_\alpha = \frac{A_0}{16}$ $A_\beta = \frac{A_0}{4}$ $A_d = \frac{A_0}{2}$ $\therefore Atotal = \frac{A_0}{16} + \frac{A_0}{4} + \frac{A_0}{2} = \frac{A_0 + 4A_0 + 8A_0}{16}$

$\therefore Atotal = \frac{13A_0}{16}$ $P\alpha = \frac{A_1\alpha}{A_{total}} = \frac{1}{13}$ $\therefore n = 1$

36. $t_{h1} = 3years$ $N_1 = 2N$

$t_{h2} = 2years$ $N_2 = N$ $\frac{N_0}{2N} = (2)^{\frac{x}{3}}$ $\frac{N_0}{N} = (2)^{\frac{x}{2}}$

37. At maximum extension along line joining $v = 0$

But in $\perp v$ direction remain same $\frac{1}{2}mv^2\times 2 = \frac{1}{2}k\eta^2 + \frac{1}{2}m\left(\frac{v}{\sqrt{2}}\right)^2\times 2$

38. $I_{requined} = I_{Total\,disc} - I_{cnt\,disc}$

39. $\vec{F} = q\left(\vec{V}\times\vec{B}\right) and\ \vec{F} = q\vec{E}$

40. **(P)** $y = \frac{\omega^2x^2}{2g} = 1.25$ (R) $y = 5\tan\theta = \frac{5a}{g} = 2.5m$

(Q) $y = \frac{w^2\times 2}{2g} = 5$ (S) Equating pressure at junction $\tan\theta = \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1}$

MATHEMATICS

41. Rearranging the given relation gives us

$$\left(1+\frac{b}{a}\right)\sin^4\alpha+\left(1+\frac{a}{b}\right)\cos^4\alpha=1$$

$$\Rightarrow \sin^4\alpha+\cos^4\alpha+\frac{b}{a}\sin^4\alpha+\frac{a}{b}\cos^4\alpha=1$$

Since $\sin^4\alpha+\cos^4\alpha=\left(\sin^2\alpha+\cos^2\alpha\right)^2-2\sin^2\cos^2\alpha=1-2\sin^2\alpha\cos^2\alpha$, we have

$$\Rightarrow \frac{b}{a}\sin^4\alpha+\frac{a}{b}\cos^4\alpha-2\sin^2\alpha\cos^2\alpha=0$$

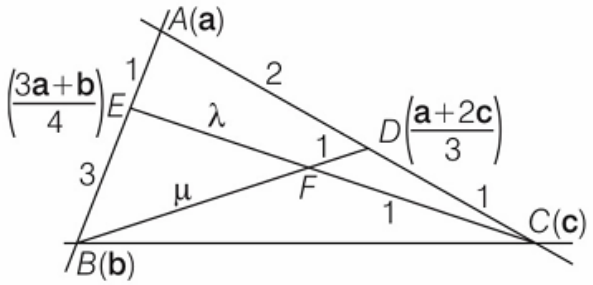
$$\Rightarrow \left(\sqrt{\frac{b}{a}}\sin^2\alpha-\sqrt{\frac{a}{b}}\cos^2\alpha\right)^2=0$$

$$\Rightarrow \frac{\sin^2\alpha}{a}=\frac{\cos^2\alpha}{b}=\frac{\sin^2\alpha+\cos^2\alpha}{a+b}=\frac{1}{a+b}$$

$$\Rightarrow \sin^2\alpha=\frac{a}{a+b},\cos^2\alpha=\frac{b}{a+b}$$

$$\Rightarrow \frac{\sin^2\alpha}{a^3}+\frac{\cos^8\alpha}{b^3}=\frac{a}{(a+b)^4}+\frac{b}{(a+b)^4}=\frac{1}{(a+b)^3}$$

42. Positive vector of point $E=\dfrac{3a+b}{4}$



Position vector of point $D=\dfrac{a+2c}{3}$

Let point F divides EC in $\lambda:1 \text{ and } BD \text{ in } \alpha:1$

Then $\dfrac{b+\frac{\alpha}{3}(a+2c)}{\alpha+1}=\dfrac{\lambda c+\frac{3a+b}{4}}{\lambda+1}$

$$\left[b+\frac{\alpha}{3}(a+2c)\right](\lambda+1)=\left(\lambda c+\frac{3a+b}{4}\right)(\alpha+1)$$

Comparing the coefficient of a, b and c

$$\Rightarrow \frac{\infty(\lambda+1)}{3}=\frac{3(\alpha+1)}{4} \qquad \ldots. \text{(i)}$$

$$\Rightarrow \lambda+1=\frac{\alpha+1}{4} \qquad \ldots. \text{(ii)}$$

$$\frac{2\alpha}{3}(\lambda+1)=\lambda(\alpha+1) \qquad \ldots\ldots \text{(iii)}$$

On solving we get $\lambda=\dfrac{3}{2}$

43. Let $y_1=12-\sqrt{1-x_1^2} \text{ and } y_2=\sqrt{4x_2} \text{ or } x_1^2+(y_1-12)^2=1 \text{ and } y_2^2=4x_2$

Thus (x_1,y_1) lies on the circle $x^2+(y-12)^2=1$ and (x_2,y_2) lie on the parabola $y^2=4x$.

Thus, given expression is the shortest distance between the curves

$x^2+(y-12)^2=1 \text{ and } y^2=4x$

Now the shortest distance always occurs along common normal to the curves and normal to circle passes through the centre of the circle

Normal to the parabola $y^2=4x$ is $y=mx-2m-m^3$ passes through $(0,12)$ gives

$m^3+2m+12=0$,

which has only one real root m=-2

Hence, corresponding point on the parabola is (4,4)

Thus, required minimum distance $=\sqrt{4^2+8^2}-1=4\sqrt{5}-1$

44. Given, $\begin{vmatrix} x^3+1 & x^2y & x^2z \\ xy^2 & y^3+1 & y^2z \\ xz^2 & yz^2 & z^3+1 \end{vmatrix}=11$, taking x,y,z common from C_1,C_2,C_3 respectively, then

$$xyz\begin{vmatrix} \frac{x^3+1}{x} & x^2 & x^2 \\ y^2 & \frac{y^3+1}{y} & y^2 \\ z^2 & z^2 & \frac{z^3+1}{z} \end{vmatrix}=11$$

On multiplying $R_1 \text{ by } x, R_2$ by y and $R_3 \text{ by } z$, we get

$$\Rightarrow \begin{vmatrix} x^3+1 & x^3 & x^3 \\ y^3 & y^3+1 & y^3 \\ z^3 & z^3 & z^3+1 \end{vmatrix}=11$$

Applying $R_1 \to R_1+R_2+R_3$, then

$$\Rightarrow \begin{vmatrix} x^3+y^3+z^3+1 & 0 & 0 \\ y^3 & 1 & 0 \\ z^3 & 0 & 1 \end{vmatrix}=11$$

$\Rightarrow x^3+y^3+z^3+1=11$

$\Rightarrow x^3+y^3+z^3=10$

Therefore, the ordered triplets are (2,1,1), (1,2,1) and (1,1,2)

45. Solving $C_1 \text{ and } C_2$ we get P as $\left(\frac{r^2}{2},\sqrt{r^2-\frac{r^4}{4}}\right)$

Solving the line PQ with x-axis we get x co-ordinate as $\alpha=\frac{\frac{r^3}{2}}{r-\sqrt{r^2-\frac{r^4}{4}}}$

$$\Rightarrow \alpha_{\lim}=\lim_{r\to 0^+}\frac{\frac{r^3}{2}}{r-\sqrt{r^2-\frac{r^4}{4}}}=4$$

R approaches the point (4,0) as $r \to 0^+$

46. $Let\ f'(x)=a\sin x+b\cos x<0 \forall x\in\left[\pi,\frac{3\pi}{2}\right]$

$\Rightarrow F(x)$ is decreasing in $\left[\frac{7\pi}{6}, \sqrt{21}\right]$

So, F(x) is maximum at $x = \frac{7\pi}{6}$ which is 0.

47. Assume O to be the origin, and the position vectors of A, B, C, D to be $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively.

Since $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are non-coplanar vectors, we must have scalars $\lambda_1, \lambda_2, \lambda_3, \lambda_4$, such that $\lambda_1\vec{a} + \lambda_2\vec{b} + \lambda_3\vec{c} + \lambda_4\vec{d} = \vec{0}$

Since AO is produced to meet the plane of the opposite face in P, $\overrightarrow{AO}$ and $\overrightarrow{OP}$ must be collinear vectors. Thus,

$\overrightarrow{AO} = \mu\overrightarrow{OP}$ for some $\mu \in \mathbb{R}$

$\Rightarrow -\vec{a} = \mu\overrightarrow{OP} = \mu\vec{p}$ ($\vec{p}$ is the position vector of P)

This when used in (1) gives

$(-\mu\lambda_1)\vec{p} + \lambda_2\vec{b} + \lambda_3\vec{c} + \lambda_4\vec{d} = \vec{0}$

However, Since B,C,D and P will be coplanar, we have

$-\mu\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4 = 0 \Rightarrow \mu = \frac{\lambda_1}{\lambda_2 + \lambda_3 + \lambda_4}$

$\Rightarrow \frac{OP}{AP} = \frac{\left|\overrightarrow{OP}\right|}{\left|\overrightarrow{AP}\right|} = \frac{\mu}{1+\mu} = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3 + \lambda_4} = \frac{\lambda_1}{\sum\lambda_i}$

We similarly have, $\frac{OQ}{BQ} = \frac{\lambda_2}{\sum\lambda_i}, \frac{OR}{CR} = \frac{\lambda_3}{\sum\lambda_i}, \frac{OS}{DS} = \frac{\lambda_4}{\sum\lambda_i} \Rightarrow \frac{OP}{AP} + \frac{OQ}{BQ} + \frac{OR}{CR} + \frac{OS}{DS} = 1$

48. $$f(x) = \begin{cases} \left\{\log_a\left(a^{[x]+[-x]}\right)^x\right\}\left(\dfrac{a^{\left(\frac{[x]+[-x]}{|x|}\right)^{-5}}}{3+a^{\frac{1}{|x|}}}\right); & |x| \neq 0; a > 1 \\ 0 & x = 0 \end{cases}$$

$$f'(0) = \lim_{h\to 0}\frac{f(h+0) - f(0)}{h}$$

$$= \lim_{h\to 0^+}\frac{\left\{\log_a\left(a^{[h]+[-h]}\right)^h\right\}.\left(\dfrac{a^{\left(\frac{[h]+[-h]}{|h|}\right)^{-5}}}{3+a^{\frac{1}{|h|}}} - 0\right)}{h}$$

$$=\begin{cases}\lim\limits_{h\to 0^-}\dfrac{\left\{\log_a\left(a\left|[h]\right|+[-h]^h\right)\right\}.\left(\dfrac{a^{\frac{2}{\frac{[h]+[-h]}{|h|}}-5}}{3+a^{\frac{1}{|h|}}}\right)}{h}\\ \lim\limits_{h\to 0^+}\dfrac{\left\{\log_a\left(a\left|[h]\right|+[-h]^h\right)\right\}.\left(\dfrac{a^{\frac{2}{\frac{[h]+[-h]}{|h|}}-5}}{3+a^{\frac{1}{|h|}}}\right)}{h}\end{cases}$$

$$=\begin{cases}\lim\limits_{h\to 0^-}\dfrac{\left\{h.\log_a\left(a\left|-1+0\right|\right)\right\}.\left(\dfrac{a^{\frac{2}{\frac{-1+0}{h}}-5}}{3+a^{-1/h}}\right)}{h}\\ \lim\limits_{h\to 0^+}\dfrac{\left\{h.\log_a\left(a\left|-1+0\right|\right)\right\}.\left(\dfrac{a^{\frac{2}{\frac{0-1}{h}}-5}}{3+a^{1/h}}\right)}{h}\end{cases}$$

$$=\begin{cases}\lim\limits_{h\to 0^-}\left\{\log_a a\right\}\left(\dfrac{a^{2h-5}}{3+a^{-1/h}}\right)\\ \lim\limits_{h\to 0^+}\left\{\log_a a\right\}\left(\dfrac{a^{(-2h-5)}}{3+a^{1/h}}\right)\end{cases}=\begin{cases}\lim\limits_{h\to 0^-}\dfrac{a^{2h-5}}{3+a^{-1/h}}\\ \lim\limits_{h\to 0^+}\dfrac{a^{-2h-5}}{3+a^{-1/h}}\end{cases}$$

$$\begin{cases}0 & \left(\because\left(\lim\limits_{h\to 0^-}\left(3+a^{-1/h}\right)=\infty\right)\right.\\ 0 & \left.\left(\lim\limits_{h\to 0^+}\left(3+a^{1/h}\right)=\infty\right)\right)\end{cases}$$

$\therefore f'(0)=f'\left(0^-\right)=f'(0')=0$

Hence f(x) is differentiable at x=0 and also continuous at x=0

49. As maximum value of $\dfrac{x^2-2x+d}{x^2+3x+d}$ is 6

$\Rightarrow \dfrac{x^2-2x+d}{x^2+3x+d}=6$ has equal roots

$\Rightarrow 5x^2+20x+5d=0 \Rightarrow 16-4d=0 \Rightarrow d=4 \,and\, a=\dfrac{2}{7}$

50. $I=\int\limits_0^\infty \dfrac{\ln t}{x^4+t^2}dt; let\, t=x^2\tan\theta \Rightarrow dt=x^2\sec^2\theta d\theta$

$\Rightarrow I=2\int\limits_0^{\pi/2}\dfrac{\ln\left(x^2\tan\theta\right)x^2\sec^2\theta}{x^4\sec^2\theta}d\theta=\dfrac{\pi}{2}\ln 2$

$\Rightarrow \dfrac{2\ln x^2}{x^2}.\dfrac{\pi}{2}=\dfrac{\pi}{2}\ln 2 \Rightarrow \dfrac{\ln x^2}{x^2}=\dfrac{\ln 2}{2}\Rightarrow x^2=4\,or\,2$

$\Rightarrow x=\pm 2;\pm\sqrt{2}$

51. Let the point on the lien be $(2\lambda, 2+2\lambda, 1)$

Now as it strikes $x=0 \Rightarrow \lambda = 0 \Rightarrow A(0,2,1)$

Find the reflection of any point P on the line along x=0 let it be Q then AQ will be reflected ray.

52. $\because S(k,n) = \sum_{r=0}^{n} r^2.\left({}^{n}C_r\right)a^r.b^{n-r} \qquad = b^n \sum_{r=0}^{n} r^2.\left({}^{\frac{n}{r}.n-1}C_{r-1}\right).\left(\frac{a}{b}\right)^r$

$= nb^n \sum_{r=0}^{n}\left((r-1)+1\right)^{n-1} C_{r-1}.\left(\frac{a}{b}\right)^r \qquad = nb^n \sum_{r=0}^{n}\left((n-1).^{n-2}C_{r-1}\right)\left(\frac{a}{b}\right)^r$

$= nb^n.(n-1).\left(\frac{a}{b}\right)^2 \sum_{r=0}^{n} {}^{n-2}C_{r-2}\left(\frac{a}{b}\right)^{r-2} + nb^n.\left(\frac{a}{b}\right)\sum_{r=0}^{n} {}^{n-1}C_{r-}\left(\frac{a}{b}\right)^{r-1}$

$= nb^n.(n-1)\left(\frac{a}{b}\right)^2\left(1+\frac{a}{b}\right)^{n-2} + nb^n.\left(\frac{a}{b}\right).\left(1+\frac{a}{b}\right)^{n-1}$

$= n(n-1)a^2k^{n-2} + nak^{n-1}$

$= n^2a^2k^{n-2} + nak^{n-2}(k-a) = n^2a^2k^{n-2} + nabk^{n-2}$

$\therefore S(1,3)9a^2 + 3ab = 3\left(3a^2+ab\right) \qquad [\because a+b=k]$

$S(2,4) = 16\left(4a^2+ab\right) \qquad S(3,5) = 135\left(5a^2+ab\right)\ S(4,6) = 1536\left(6a^2+ab\right)$

53. $\omega = \cos 12^o + \cos 48^o + i\left(\cos 48^o + \sin 12^o\right) = 2\cos 30^o \cos 18^o + 2i\sin 30^o \cos 18^o = 2\cos 18^o\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$

$z = \omega^6 = 2^6 \cos^6 18^o\left(\cos\pi + i\sin\pi\right)$

$z = -2^6\left(\cos^6 18^o\right) \qquad I_m(z) = 0$

54. Let $x_1, x_2, x_3, ..., x_7$ person are entering into gate A,B,C,D,….. G gas respectively

Now, $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 = 200$

Total no. of ways to enter: ${}^{200+7-1}C_{7-1} = {}^{206}C_6 = {}^{206}C_{200}$

Now the arrangement of these 200 persons will also count so

Total no. of ways $= {}^{206}C_{200} \times \underline{|200} = {}^{206}P_{200}$

n=206, r=200

n-r=6

55. We know, $\sin 3\theta = 3\sin\theta - 4\sin^3\theta = \sin\theta(1+2\cos 2\theta) \qquad \therefore \sin 3\theta = \sin\theta(1+2\cos 2\theta)$ …. (1)

Now, on putting $3\theta = x, \frac{x}{3}, \frac{x}{3^2}, ...$ one by one in Eq. (i), we get $\Rightarrow \sin x = \sin\left(\frac{x}{3}\right).\left(1+2\cos\frac{2x}{3}\right)$

$$\Rightarrow \frac{\sin x}{x} = \lim_{n\to\infty}\frac{\sin x/3^n}{x/3^n}\prod_{k=1}^{n}\frac{\left(1+2\cos\left(\frac{2x}{3^k}\right)\right)}{3^n} \Rightarrow \frac{\sin x}{x} = \prod_{k=1}^{\infty}\left(\frac{1+2\cos\left(\frac{2x}{3^k}\right)}{3}\right) = f(x)$$

Thus, $|xf(x)| + \|x-2|-1| = \frac{|x||\sin x|}{|x|} + \|x-2|-1|$ is not

Differentiable at $\{\pi, 2\pi, 1, 2, 3\}$ number of points are 5

56. Let E_1, E_2, E_3, E_4 be the events that first two drawn book are (math, math) (math, phy) (phy, math) and (phy, phy) respectively and A be the event that third drawn book is of math.

Here $P(E_1) = \frac{3}{5}\times\frac{5}{7} = \frac{3}{7}, P(E_2) = \frac{3}{5}\times\frac{2}{7} = \frac{6}{35}, P(E_3) = \frac{2}{5}\times\frac{3}{4} = \frac{3}{10}, P(E_4) = \frac{2}{5}\times\frac{1}{4} = \frac{1}{10}$

Also, $P(A/E_1)=\frac{7}{9}; P(A/E_2)=5/6; P(A/E_3)=\frac{5}{6}; P(A/E_4)=1$

Now by Bay's theorem $P(E_4/A)=\frac{P(E_4).P(A/E_4)}{P(A)}$

$$=\frac{\frac{1}{10}\times 1}{\frac{3}{7}\times\frac{7}{9}+\frac{6}{35}\times\frac{5}{6}+\frac{3}{10}\times\frac{5}{6}+\frac{1}{10}\times 1}=\frac{1}{\frac{10}{3}+\frac{10}{7}+\frac{5}{2}+1}=\frac{42}{140+60+105+42}=\frac{42}{347}$$

57. Conceptual

58. $4x+7y+4z+81=0$ ……. (i)

$5x+3y+10z=25$ …….. (ii)

Equation of plane of passing through their line of intersection is

$(4x+7y+4z+81)+\lambda(5x+3y+10z-25)=0$

Or $(4+5\lambda)x+(7+3\lambda)y+(4+10\lambda)z+81-25\lambda=0$ ….. (iii)

Plane (iii) is $\perp$ to (i) so

$4(4+5\lambda)+7(7+3\lambda)+4(4+10\lambda)=0 \quad \therefore \lambda=-1$

From (iii) equation of plane is $-z+4y-6z+106=0$ …. (iv)

Distance of (iv) from (0,0,0) $=\frac{106}{\sqrt{1+16+36}}\frac{106}{\sqrt{53}}$

59. g(x) can be defined is

$$g(x)=\begin{cases} f(x), & -2\le x<-1 \\ -1, & -1\le x<0 \\ 0, & 0\le x\le 2 \\ f(x), & 2\le x\le 3 \end{cases} \text{ or } g(x)=\begin{cases} x^2+2x, & -2\le x<-1 \\ -1, & -1\le x<0 \\ 0, & 0\le x\le 2 \\ x^2-2x & 2\le x\le 3 \end{cases}$$

Clearly g(x) is discontinuous at x=0, g(x) is not differentiable at x=0,2

60. C_1+C_2 is decreasing function and its domain will be $[-1,1]$

